Screw It! README

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Target Audience

Screw It! aims to assist renters and homeowners beginning to learn about home maintenance. The value proposition of Screw It! is that it's to be a central hub for home maintenance help, and makes home maintenance decision-making easy, through DIY (Do It Yourself) vs. professional services comparisons.

Access & Setup

You can access the Screw It! Medium-Fi Prototype on Figma using this link.
https://www.figma.com/proto/IMQ2zVDXd33DWxsgweU8iZ/Screw-It-Medium-Prototype?node-id=0-1&t=vpS2D4JpvMNykql7-1
The prototype should be ready to use once you click on the link and the page is loaded.

Intended Device

Screw It! is meant to be run on mobile smartphone devices such as Apple's iPhone, Samsung's Galaxy, or Google's Pixel. However, a desktop computer or laptop (Mac,

Windows, Linux, etc.) is recommended for a user to access the prototype to view it on a larger screen.

Context

Screw It!'s purpose is to assist homeowners and renters to learn and identify home maintenance issues in their household. The user should be able to accomplish the following:

Identify

Screw It! helps homeowners and renters identify a home maintenance issue by uploading or taking a picture. With the help of our AI assistant, Angie, the AI chatbot informs the user what is seen, the likely cause, and what could be damaged. The user then has the option to compare solutions, be given step-by-step instructions on how to fix the issue, or find a professional.

Compare

Depending on the given issue, Screw It! gives homeowners the option to compare DIY (Do It Yourself) and Professional services on the basis of estimated cost and time, with a description explaining each case. If the user decides to DIY (Do It Yourself), they are given step-by-step instructions. If the user decides to pick a professional, they are given a list of nearby professionals around their current location.

Learn

Depending on the given issue, Screw It! gives a DIY (Do It Yourself) overview before showing a step-by-step video tutorial with written instructions. The user is also given the option to use AR (Augmented Reality) and their camera to have the AI assistant give live guidance and mark where the issue lies.

Prototype Operation

General

With Figma, some of our features are supported. In addition, through some screens, users have the option to speak to Angie, our AI assistant chatbot. When clicking

each screen, the features highlighted in blue are interactable. Any other feature is not clickable and has not been implemented in this Medium-Fi prototype.

Simple Task (Identify a home maintenance problem)

- From the home screen, the user can click the logo to go to our home screen
- The user can click the yellow arrow beside the label "Identify Issue."
- The user is given the option to take a picture or upload a picture of their home maintenance issue (a spraying water faucet)
- After a loading screen, Angie, our AI assistant chatbot, states what they see from the picture and identifies a likely issue as a "Broken Aerator."
- The user is given the choice to compare DIY (Do It Yourself) and professional services, give step-by-step instructions, to find a professional, or go back to the home screen.
- If the user needs more assistance, they have the option to ask Angie more questions through a chat.

Moderate Task (Compare the DIY and professional solutions)

- From the home screen, the user can click the logo to go to our home screen
- The user can click the yellow arrow beside the label "Compare Solutions."
- The user is given the option to search or upload a picture of their home maintenance issue (Broken aerator), or search through given popular issues or the user's recent searches
- The user inserts and searches for "broken aerator."
- The user is told the complexity of fixing a broken aerator. Beneath, there are two options, "DIY (Do It Yourself)" or "Professional", and a quick summary describing the estimated cost and time.
- If the user clicks on "DIY (Do It Yourself) or "Professional," the user is given a longer explanation of how a homeowner or professional would solve the problem.
    - If the user is on the "DIY (Do It Yourself)" page, they have the option to view "Step by Step Instructions" on how to fix the problem
    - If the user is on the "Professional" page, they have the option to "Find a Professional" near their current location.
- If the user needs more assistance, they have the option to ask Angie more questions through a chat.

Complex Task (Learn how to fix the home maintenance problem)

- From the home screen, the user can click the logo to go to our home screen
- The user can click the yellow arrow beside the label "Compare Solutions."
- The user is given the option to search or upload a picture of their home maintenance issue (Broken aerator), or search through given popular issues or the user's recent searches
- The user inserts and searches for "broken aerator."
- The user is given a short description of the task and what it entails, such as how long it would take, the process, and the intended results.
- After the user clicks the "next" button, the user is given a video tutorial with written instructions for each step on how to solve the problem. In this prototype, there are 3 steps, and in each step, the user has options to go back a step, to move on to the next step, or to use AR (Augmented Reality).
- If the user chooses to click on "AR" (Augmented Reality), the user can move their camera around for clearer guidance, where Angie, the AI assistant, can clearly identify and locate specific objects needed for the current step.
- If the user needs more assistance, they have the option to ask Angie more questions through a chat.

Prototyping Tools

The prototype was created using Figma for the design of the prototype and wireframing. Some images were taken by us (i.e. the spraying faucet) and were taken from Google. The Screw and AI Chatbot logo was designed by us. All other icons, buttons, etc. were created within Figma.

Limitations

The following list includes some limitations that occurred with our medium-fi prototype on Figma:

- The users cannot test issues other than the "broken aerator."
- The AI chatbot and AR feature are not actually implemented and cannot be tested.
    - We do not know how long the AI can identify the issue, and we cannot guarantee the accuracy of the AI assistant.

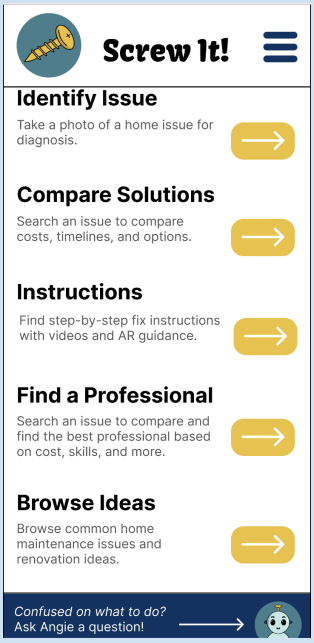
- The users cannot test the video playback or find a relevant video
- The user cannot test how long the loading screens would actually take
- The user cannot test calling professional services

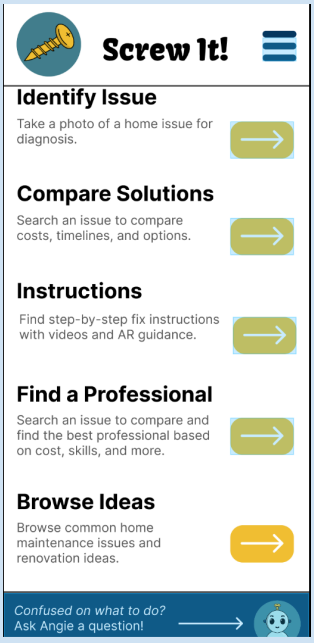
Wizard of Oz Techniques/Hard-Coded Items

Due to Figma's limitations, we added Wizard of Oz techniques and hard-coded items, including:

- The issue is always with a broken aerator
    - The search function will always input "broken aerator."
- The video shows screenshots of a broken aerator
- AR is just a photo and cannot show moving around and 3D features
- AI doesn't respond
- The professional services and reviews are not real and are made up
- When a picture is uploaded, the loading sequence is fake and will automatically switch to the next screen after 500ms

Images





(Features highlighted in blue indicate that the feature is interactable. Non-highlighted features mean that the feature has not been implemented.)